

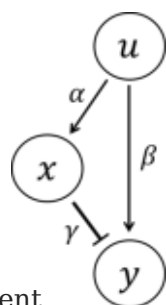
A Incoherent Feedforward Loop (IFFL)

$$\frac{dx}{dt} = \alpha u - \delta x$$

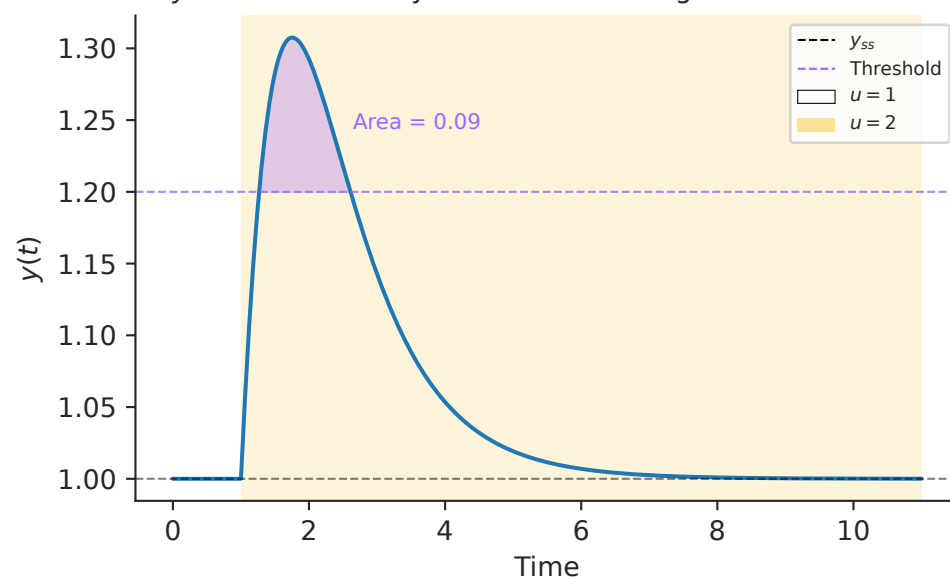
$$\frac{dy}{dt} = \beta u - \gamma x y$$

Steady state: $y_{ss} = \frac{\beta \delta}{\gamma \alpha}$

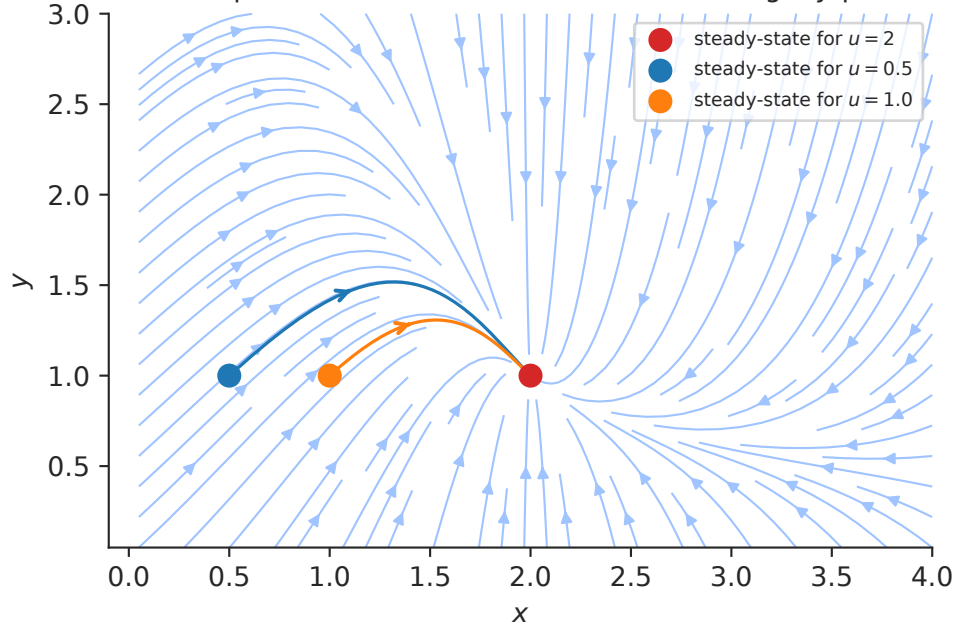
The output's steady state is independent of the input u , making it robust to changes in u .



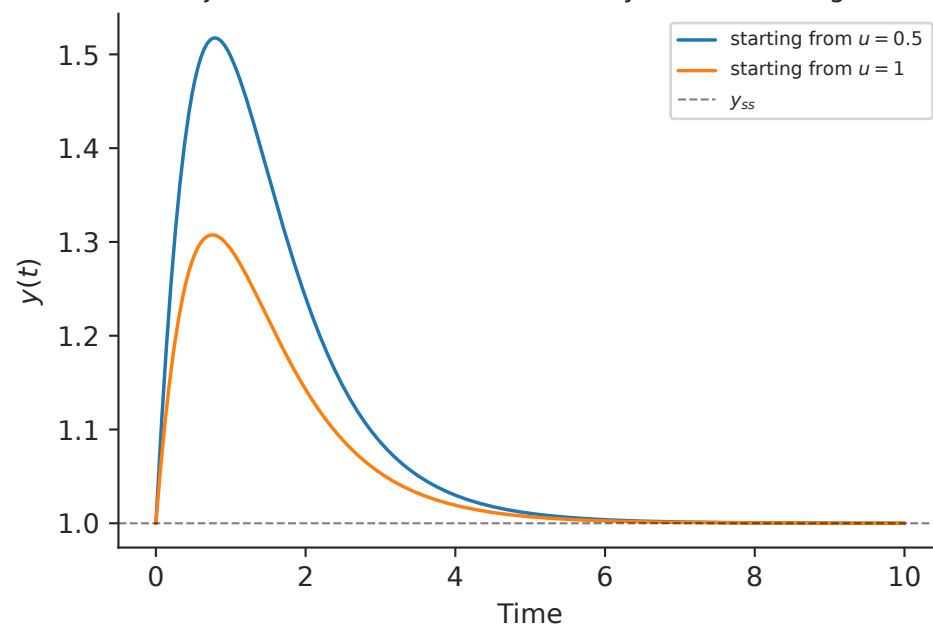
B y returns to steady-state after u changes



C Phase portrait for u = 2. Smaller x causes a larger y pulse

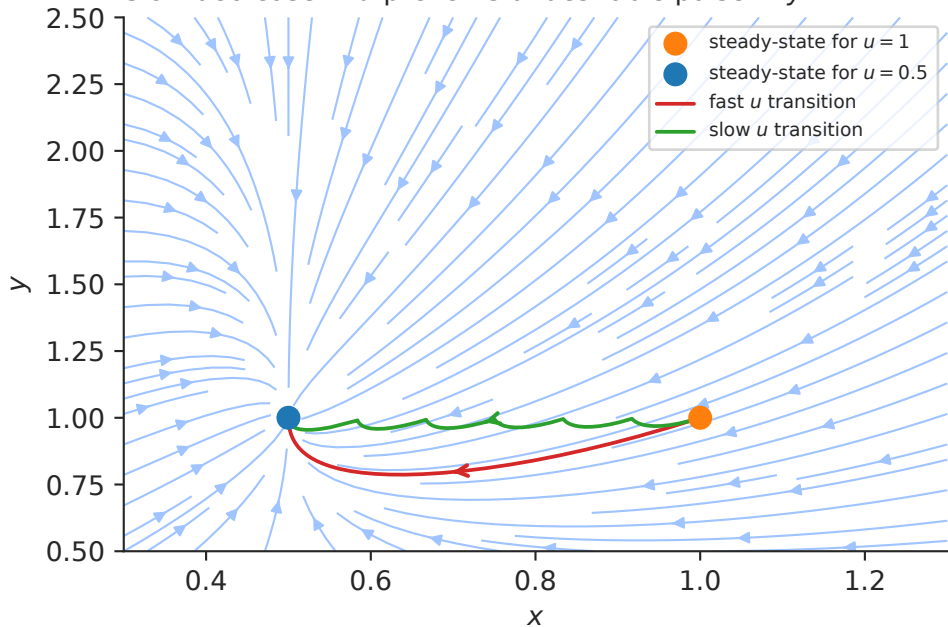


C' Trajectories over time, immediately after switching to u = 2

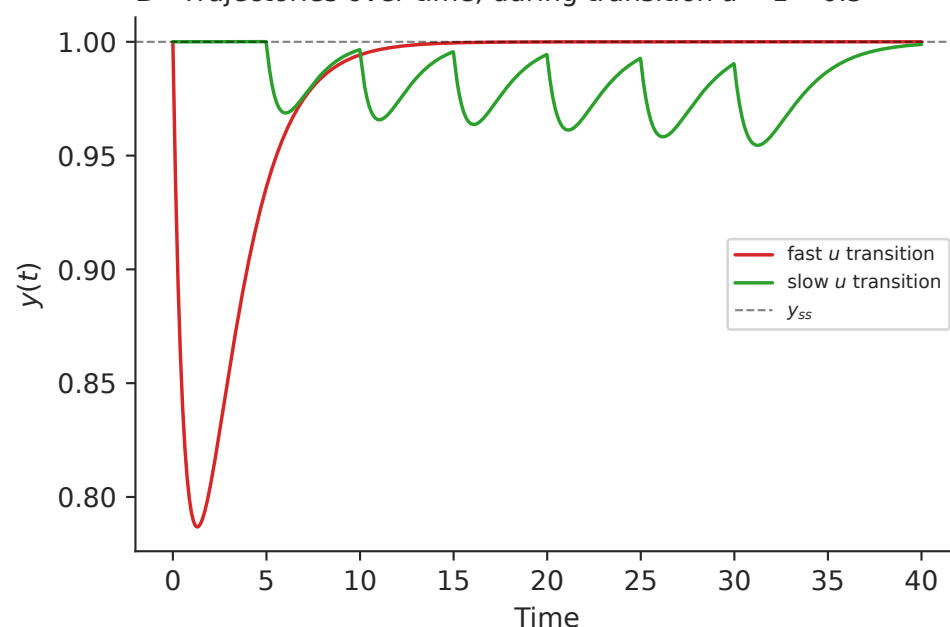


D Phase portrait for u = 0.5

Slow decrease in u prevents undesirable pulse in y



D' Trajectories over time, during transition u = 1 → 0.5



E Adaptive control achieves a larger peak for output y

